

A lifetime distribution with decreasing failure rate

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Abstract

A two-parameter distribution with decreasing failure rate is introduced. Various properties are discussed and the estimation of parameters is studied by the method of maximum likelihood. The estimates are attained by the EM algorithm and expressions for their asymptotic variances and covariances are obtained. Numerical examples based on real data are presented. © 1998 Elsevier Science B.V. All rights reserved

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1. Introduction

The study of length of life of organisms, devices, structures, materials, etc., is of major importance in the biological and engineering sciences. A substantial part of such study is devoted to the mathematical description of the length of life by a failure distribution. Sometimes physical considerations of the failure mechanism may lead to a specific distribution but more often, the choice is made on the basis of how well the actual observations of times to failure appear to be fitted by the distribution. In reliability studies a useful tool for reducing the range of possible candidates is provided by the shape and monotonicity of the failure rate function since it reflects some of the characteristics of the mechanism leading to conclusion of life.

Situations where the failure rate function decreases with time have been reported by several authors. Indicative examples are the business mortality (Lomax, 1954), failures in the air-conditioning equipment of a fleet of Boeing 720 aircrafts or in semiconductors from various lots combined (Proschan, 1963) and the life of integrated circuit modules (Saunders and Myhre, 1983). In general, a population is expected to exhibit decreasing failure rate (DFR) when its behavior over time is characterized by ‘work hardening’ (in engineering terms) or ‘immunity’ (in biological terms); sometimes the broader term ‘infant mortality’ is used to denote the DFR phenomenon. The resulting improvement of reliability with time might have occurred by means of actual physical changes that caused self-improvement or simply it might have been due to population heterogeneity. Indeed, Proschan (1963) proved that the DFR property is inherent to mixtures of distributions

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with constant failure rate (see also McNolty et al., 1980 for other properties of exponential mixtures) and Gleser (1989) demonstrated the converse for any gamma distribution with shape parameter less than one. In addition, Gurland and Sethuraman (1994) give examples illustrating that such results may hold for mixtures of distributions with rapidly increasing failure rate. Under the same theory of heterogeneity Cozzolino (1968) developed various DFR models of practical importance. General theoretical aspects of DFR distributions concerning properties and estimation techniques from complete and censored samples were explored by Barlow et al. (1963), Barlow and Marshall (1964, 1965), Marshall and Proschan (1965) and Saunders and Myhre (1983).

In this article we present a new DFR distribution that fits well real data and can be a promising candidate for a model of such lifetime behavior. We study various of its properties and explore relations with some known distributions. The estimation of parameters is considered by the method of maximum likelihood. We carry out an investigation of the ml equations and derive an EM algorithm for the computation of the parameter estimates. The observed and expected Fisher informations of the parameters are also obtained. Numerical examples using real data are included.

2. The distribution

Suppose that $\{Y_i\}_{i=1}^Z$ are i.i.d. with density $f(y; \beta) = \beta e^{-\beta y}$, for $\beta, y \in \mathbb{R}_+$ and Z is a geometric variable with probability function $P(z; p) = (1 - p)p^{z-1}$, for $z \in \mathbb{N}$ and $p \in (0, 1)$. Let $X = \min(\{Y_i\}_{i=1}^Z)$. Then $f(x|z; \beta) = \beta z e^{-\beta z x}$ and

$$f(x; \theta) = \beta(1 - p)e^{-\beta x}(1 - pe^{-\beta x})^{-2} \quad (2.1)$$

with $\theta = (\beta, p)$. The latter defines the distribution that we shall be referring to in the sequel as the exponential-geometric distribution (EG for brevity). In the literature it is customary for such names to be given to distributions arising via the operation of compounding (mixing). Indeed, the EG distribution can also be obtained by compounding an exponential with a geometric distribution. Specifically, it is easily seen that by assuming that X follows an exponential distribution with parameter γ and by taking γ to behave as βz , where Z is a geometric variable with parameter p , then the distribution of X is the EG with density given by (2.1).

To attach an interpretation to the proposed model, we may think of a situation where failure (of a device for example) occurs due to the presence of an unknown number, Z , of initial defects of the same kind (a number of semiconductors from a defective lot, for example). The Y 's represent their lifetimes and each defect can be detected only after causing failure, in which case it is repaired perfectly. Then the distributional assumptions given earlier lead to the EG distribution for modeling the time to the first failure.

The mode of the distribution is at zero and the modal value is $\beta(1 - p)^{-1}$. For all values of the parameters the density is strictly decreasing in x and tends to zero as $x \rightarrow \infty$. Its graph resembles those of the Pareto II and exponential distributions. As p approaches zero the EG tends to the exponential distribution with parameter β .

3. Properties of the distribution

3.1. Relations with other distributions

It can be verified by standard techniques that if X is an EG variable, with density given by (2.1), then

- (i) the random variable $Y = p^{-1}(e^{\beta X} - 1)$ follows the Pareto II distribution (also known as the Lomax distribution) with shape and scale parameters one and $p/(1 - p)$, respectively,

(ii) the random variable $Y = -\beta^{-1} \ln(1 - e^{-\beta X})$ follows the logistic distribution, truncated in $\mathbb{R}_+$, with location and scale parameters $\beta^{-1} \ln\{p(1-p)^{-1}\}$ and β , respectively.

Relations with other distributions can be established through those in (i) and (ii). For example, in (ii) $Y_1 = \exp Y = (1 - e^{-\beta X})^{-1/\beta}$ has the log-logistic distribution, truncated in $(1, \infty)$, with the same location and scale parameters and if $p = 2^{-1}$ then Y in (i) is a beta-prime distribution so that further relations with, say, the F the extreme value and chi-squared distributions are readily available (see Patil et al., 1984, pp. 24, 91 and 117).

Furthermore, if the common distribution of the Y 's, in Section 2, is replaced by the EG with parameters β, η then the distribution of X is EG with parameters β and $1 - (1 - \eta)(1 - p)$.

3.2. Probabilities and moments

By straightforward integration we find that the distribution function is given by

$$F(x; \theta) = P(X \leq x) = (1 - e^{-\beta x})(1 - pe^{-\beta x})^{-1} \quad (3.1)$$

and hence the median is calculated to be $\beta^{-1} \ln(2 - p)$.

The raw moments of X may be determined from (2.1) by direct integration. For $r \in \mathbb{N}$, we find that

$$E(X^r; \theta) = (1 - p)r!(\beta^r p)^{-1}L(p; r),$$

where $L(p; r) = \sum_{j=1}^{\infty} p^j j^{-r}$ is the generalization of Euler's dilogarithm function of p , $L(p; 2)$ (see, Erdelyi et al., 1953, p. 31) and is also known as the polylogarithm function; the function is quickly evaluated and readily available in standard software such as Mathematica. The mean and variance of the EG distribution are given by

$$E(X; \theta) = \frac{(p-1)\ln(1-p)}{\beta p}, \quad \text{Var}(X; \theta) = \frac{1-p}{p\beta^2} \left\{ 2L(p; 2) - \left(\frac{1-p}{p} \right) \ln^2(1-p) \right\}. \quad (3.2)$$

Consequently, the coefficient of variation is $- \{2p(1-p)^{-1}L(p; 2) - \ln^2(1-p)\}^{1/2} \ln^{-1}(1-p)$.

3.3. The survivor, failure rate and mean residual life functions

Using (3.1), the survival probability (also known as reliability) is given by

$$s(x; \theta) \equiv 1 - F(x; \theta) = (1 - p)e^{-\beta x}(1 - pe^{-\beta x})^{-1}$$

and thus the failure rate (also known as hazard rate) for the EG distribution is

$$h(x; \theta) \equiv f(x; \theta)\{s(x; \theta)\}^{-1} = \beta(1 - pe^{-\beta x})^{-1}.$$

Both functions have simple forms (in contrast to those of, say, the gamma distribution with DFR) and this is particularly desirable when the estimation is based on data with a paucity of failures. It is easy to see that the failure rate function is decreasing although the DFR property follows from the results of Barlow et al. (1963) on mixtures. The initial and long-term hazards are $h(0; \theta) = \beta(1 - p)^{-1}$ and $h(\infty; \theta) = \beta$. They are both finite and this is conceptually appealing in contrast with, say, the decreasing Weibulls where $h(0; \theta) = \infty$ and $h(\infty; \theta) = 0$.

Given survival to time x_0 , the residual life is the period from time x_0 until the time of failure. The mean residual lifetime is

$$m(x_0; \theta) \equiv E(X - x_0 | X \geq x_0; \theta) = -(\beta p)^{-1} e^{\beta x_0} (1 - pe^{-\beta x_0}) \ln(1 - pe^{-\beta x_0}).$$

Because the EG is a DFR distribution $m(x_0; \theta)$ increases monotonically up to its limit at infinity. Clearly, $m(x_0; \theta) \rightarrow E(X; \theta)$ as $x_0 \rightarrow 0$. By applying L'Hospital's rule we find that the value of the expected residual life at infinity is β^{-1} and, consequently, that $\lim_{x_0 \rightarrow \infty} m(x_0; \theta) \{E(X; \theta)\}^{-1} = -p\{(1-p)\ln(1-p)\}^{-1} > 1$ since $p < -\ln(1-p) < p(1-p)^{-1}$ for $0 < p < 1$. Therefore, the mean residual lifetime exceeds the mean lifetime for all x_0 .

4. Estimation of parameters

4.1. Estimation by maximum likelihood

One sees from (3.2) that moment estimates for the parameters of the EG distribution with density function (2.1) cannot be obtained in closed forms and therefore there is little point in considering the method any further. Proceeding with the method of maximum likelihood, the loglikelihood function from a sample of n observations, $y_{\text{obs}} = (x_i; i = 1, \dots, n)$, is given by

$$\ell(\theta; y_{\text{obs}}) = n \ln\{\beta(1-p)\} - \beta \sum_{i=1}^n x_i - 2 \sum_{i=1}^n \ln(1 - pe^{-\beta x_i})$$

and subsequently the associated gradients are found to be

$$\begin{aligned} \frac{\partial \ell}{\partial \beta} &= n\beta^{-1} + \sum_{i=1}^n x_i - 2 \sum_{i=1}^n \{x_i(1 - pe^{-\beta x_i})^{-1}\}, \\ \frac{\partial \ell}{\partial p} &= p^{-1}\{n(p-2)(1-p)^{-1} + 2 \sum_{i=1}^n (1 - pe^{-\beta x_i})^{-1}\}. \end{aligned} \quad (4.1)$$

The solution of $\partial \ell / \partial \theta_j = 0$, $j = 1, 2$, is discussed in the following theorem.

Theorem 4.1. Let $g(\beta; p, y_{\text{obs}})$ denote the function on the RHS of the first expression in (4.1) and $\bar{x} = n^{-1} \sum_{i=1}^n x_i$. Then, for a given $p \in (0, 1)$ the root of $g(\beta; p, y_{\text{obs}}) = 0$ lies in the interval $(1-p)\{\bar{x}(1+p)\}^{-1}$ to $\bar{x}^{-1}$ and is unique.

Proof. Let $w(\beta; p, y_{\text{obs}}) = 2 \sum_{i=1}^n \{x_i(1 - pe^{-\beta x_i})^{-1}\}$. Then clearly w is strictly decreasing in β and $\lim_{\beta \rightarrow \infty} w(\beta; p, y_{\text{obs}}) = 2 \sum_{i=1}^n x_i$. It follows that

$$g(\beta; p, y_{\text{obs}}) < n\beta^{-1} + \sum_{i=1}^n x_i - \lim_{\beta \rightarrow \infty} w(\beta; p, y_{\text{obs}}) = n\beta^{-1} - \sum_{i=1}^n x_i$$

and hence $g(\beta; p, y_{\text{obs}}) < 0$ when $\beta > \bar{x}^{-1}$. On the other hand, $\lim_{\beta \rightarrow 0} w(\beta; p, y_{\text{obs}}) = 2(1-p)^{-1} \sum_{i=1}^n x_i$, so that

$$g(\beta; p, y_{\text{obs}}) > n\beta^{-1} + \sum_{i=1}^n x_i - \lim_{\beta \rightarrow 0} w(\beta; p, y_{\text{obs}}) = n\beta^{-1} - (1+p)(1-p)^{-1} \sum_{i=1}^n x_i.$$

Hence, $g(\beta; p, y_{\text{obs}}) > 0$ when $\beta < (1-p)\{\bar{x}(1+p)\}^{-1}$ and therefore there is at least one root of $g(\beta; p, y_{\text{obs}}) = 0$ in the interval $(1-p)\{\bar{x}(1+p)\}^{-1}$ to $\bar{x}^{-1}$. To prove that the root is unique we consider the first derivative of g , $g'(\beta; p, y_{\text{obs}})$, given by

$$g'(\beta; p, y_{\text{obs}}) = -n\beta^{-2} + 2p \sum_{i=1}^n \{x_i^2 e^{-\beta x_i} (1 - pe^{-\beta x_i})^{-2}\}. \quad (4.2)$$

By similar arguments one concludes that g' is not always monotonic for all $\beta > 0$. However, if β^* is a solution of $g'(\beta; p, y_{\text{obs}}) = 0$ then $n\beta^{*-1} = 2p\beta^* \sum_{i=1}^n \{x_i^2 e^{-\beta^* x_i} (1 - pe^{-\beta^* x_i})^{-2}\}$ and

$$g(\beta^*; p, y_{\text{obs}}) = \sum_{i=1}^n \frac{x_i(p^2 e^{-2\beta^* x_i} + 2p\beta^* x_i e^{-\beta^* x_i} - 1)}{(1 - pe^{-\beta^* x_i})^2}.$$

Note that the expression within parentheses is increasing in p . Hence,

$$p^2 e^{-2\beta^* x_i} + 2p\beta^* x_i e^{-\beta^* x_i} - 1 \leq 2e^{-\beta^* x_i} \{\beta^* x_i - \sinh(\beta^* x_i)\} < 0$$

for all $\beta^*, x_i > 0, i = 1, \dots, n$ and this implies that g is negative at its stationary points. In addition, $\lim_{\beta \rightarrow \infty} g(\beta; p, y_{\text{obs}}) = -\sum_{i=1}^n x_i$. It follows that g cannot have more than one root.

4.2. An EM algorithm

The maximum likelihood estimates of β and p must be derived numerically and we propose the use of the EM algorithm (Dempster et al., 1977); a statistically oriented iterative scheme with desirable properties that are fairly well known in the literature. To implement the algorithm we define the hypothetical complete-data distribution with density function $f(x, z; \theta) = \beta(1 - p)z p^{z-1} e^{-\beta z x}$, for $x, \beta \in \mathbb{R}_+, z \in \mathbb{N}, p \in (0, 1), \theta = (\beta, p)$ and assume that y_{obs} is part of the augmented data $(x_i, z_i; i = 1, \dots, n)$ from the latter distribution, resulting after failing to observe the values of Z . Under this formulation it is straightforward to verify that the E-step of an EM cycle requires the computation of the conditional expectation of $(Z|X; \theta^{(t)})$, where $\theta^{(t)} = (\beta^{(t)}, p^{(t)})$ is the current estimate of θ . Using that $P(z|x; \theta) = z(pe^{-\beta x})^{z-1}(1 - pe^{-\beta x})^2$ for $z \in \mathbb{N}$, this is found to be $E(Z|X; \theta) = 2(1 - pe^{-\beta x})^{-1} - 1$. The cycle is completed with the M-step which is essentially-full data maximum likelihood over θ , with the missing Z 's replaced by their conditional expectations. Thus, an EM iteration, taking $\theta^{(t)}$ into $\theta^{(t+1)}$, is defined by

$$\beta^{(t+1)} = n \left\{ \sum_{i=1}^n \frac{x_i(1 + p^{(t)} e^{-\beta^{(t)} x_i})}{1 - p^{(t)} e^{-\beta^{(t)} x_i}} \right\}^{-1}, \quad p^{(t+1)} = 1 - n \left(\sum_{i=1}^n \frac{1 + p^{(t)} e^{-\beta^{(t)} x_i}}{1 - p^{(t)} e^{-\beta^{(t)} x_i}} \right)^{-1}.$$

4.3. The observed and expected informations

The observed information matrix, $I(\theta; y_{\text{obs}})$, can be obtained by either differentiating (4.1) with respect to the parameters and negating the resulting expressions or by using the result of Louis (1982) (see also Meilijson, 1989) with the distribution of $Z|X$ given in Section 4.2. It is symmetric of second order with elements

$$I_{11} = -g'(\beta; p, y_{\text{obs}}), \quad I_{12} = 2 \sum_{i=1}^n \frac{x_i e^{-\beta x_i}}{(1 - pe^{-\beta x_i})^2}, \quad I_{22} = \frac{n}{(1 - p)^2} - 2 \sum_{i=1}^n \frac{e^{-\beta x_i}}{(1 - pe^{-\beta x_i})^2}$$

with g' as given in (4.2). The elements of the expected information matrix, $J(\theta)$, are calculated by taking the expectations of I_{ij} , $i, j = 1, 2$ with respect to the distribution of X , i.e. we need to find $E[X^r e^{-s\beta X} (1 - pe^{-\beta X})^{-2}]$ for $(r, s) = (2, 1), (0, 2), (1, 1)$. Expressing the relevant integrand into an infinite double series, integrating term by term and collecting the coefficients of like powers of p we obtain

$$E \left\{ \frac{X^r e^{-s\beta X}}{(1 - pe^{-\beta X})^2}; \theta \right\} = \frac{(1 - p)r!}{\beta^r p^2} \sum_{j=1}^{\infty} \binom{j+2}{3} \frac{p^{j+1}}{(s+j)^{r+1}},$$

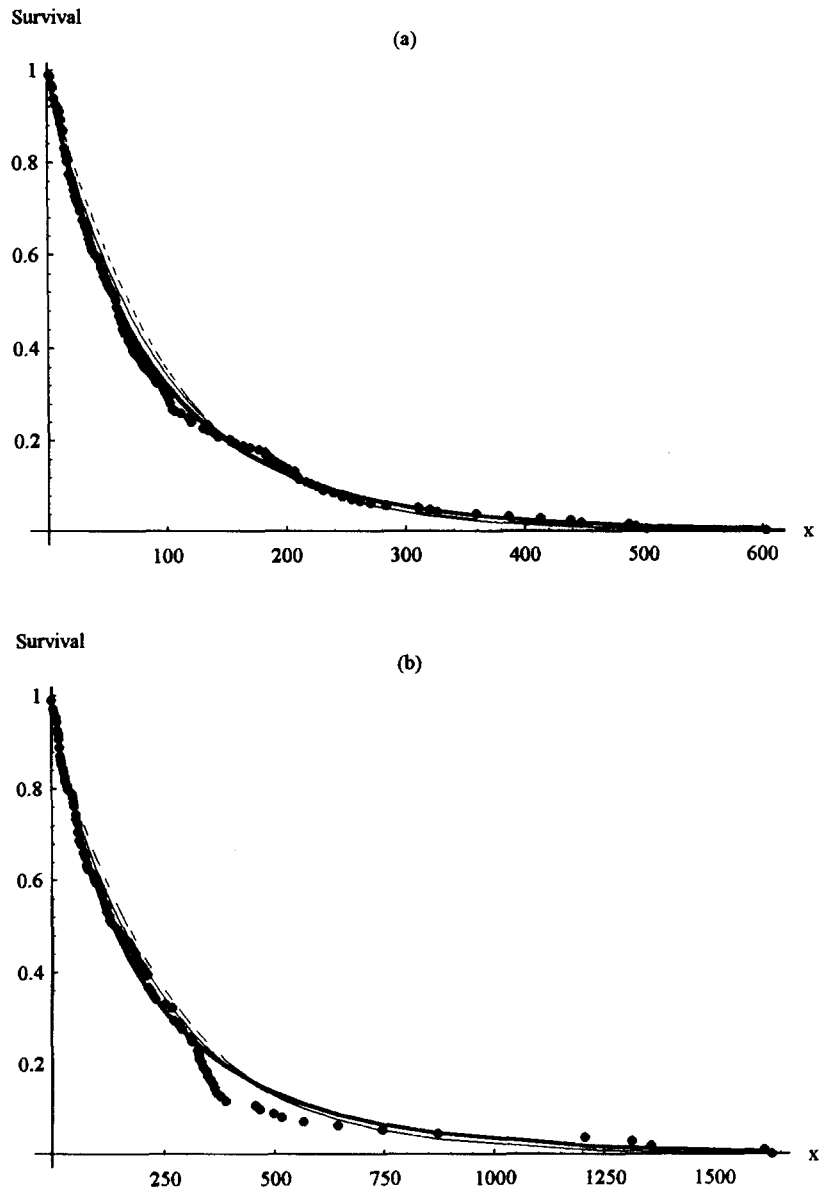


Fig. 1. Survival probability curves of the empirical distribution (dotted line), the EG distribution (thick line), the Weibull distribution (thin line) and the gamma distribution (broken line), fitted using ml estimates, for (a) the failures in the air conditioning system of a fleet of aircrafts and (b) the observations on the period between successive coal-mining disasters.

where $\binom{j+2}{3}$ denotes the usual binomial coefficient. Substituting the values of (r, s) and manipulating the elements of $J(\theta)$ are derived in the form

$$J_{11} = \frac{n}{3\beta^2 p} [3p - 2\{p - (1-p)L(p; 2)\}], \quad J_{12} = \frac{n\{4p^2 - p + (1-p)^2 \ln(1-p)\}}{3\beta p^2(1-p)},$$

$$J_{22} = n(1-p)^{-2}/3.$$

The inverse of this matrix, evaluated at $\hat{\theta}$ provides the asymptotic matrix of variances and covariances of the maximum likelihood estimates. Alternative estimates can be obtained from the inverse of the observed information matrix since it is a consistent estimator of $J^{-1}(\theta)$.

5. Examples

The fit of the EG distribution to two sets of real data, previously reported in the literature, is examined by graphical methods and the Kolmogorov–Smirnov test using maximum likelihood estimates; it is also compared with the Weibull and Gamma models with respective densities $f_1(x; \theta_1) = \beta_1 \lambda_1^{\beta_1} x^{\beta_1-1} e^{-(\lambda_1 x)^{\beta_1}}$ and $f_2(x; \theta_2) = \lambda_2^{\beta_2} x^{\beta_2-1} e^{-\lambda_2 x} \{\Gamma(\beta_2)\}^{-1}$, where $\theta_j = (\lambda_j, \beta_j)$ for $j = 1, 2$. The first set consists of the number of successive failures of the air conditioning system of each member of a fleet of 13 Boeing 720 jet airplanes. The pooled data, yielding a total of 213 observations, were first analyzed by Proschan (1963) and further discussed in Dahiya and Gurland (1972) and Gleser (1989). In the second set the data are 109 observations on the period between successive coal-mining disasters and can be found in Cox and Lewis (1978, p. 4). According to the interpretation given in Section 2, the EG model attributes the failures in the air conditioning system to the presence of defective components or errors committed during the production process and similarly, the coal-mining disasters to construction faults or human errors committed by inexperienced miners. The ml estimates $\hat{\theta} = (\hat{\beta}, \hat{p})$ and $\hat{\theta}_j = (\hat{\lambda}_j, \hat{\beta}_j)$, $j = 1, 2$ under the EG, Weibull and gamma models were, for the first set of data, $(8.01 \times 10^{-3}, 0.4276)$, $(1.12 \times 10^{-2}, 0.924)$ and $(9.9 \times 10^{-3}, 0.922)$ respectively, and for the second $(2.65 \times 10^{-3}, 0.5602)$, $(4.5 \times 10^{-3}, 0.865)$ and $(3.44 \times 10^{-3}, 0.828)$, respectively. The empirical and fitted survivor functions are plotted in Fig. 1. As the figure shows, in both cases the survival curves of the three models are very similar and they provide good fits compared with the nonparametric survivor estimates; the plots for the gamma survival curves almost overlap the Weibull curves after their crossing point, and hence these parts could not be depicted in the figure. The empirical and fitted EG survival curves are almost coincident for the first set of observations and this is also the case for most of the observations in the second set; the calculated values of the Kolmogorov–Smirnov test statistics were 0.049 and 0.076 respectively, supporting, all the same, that the EG distribution fits these data at least as good as the two popular alternatives.

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